

Walid CHIKHI

Head of NWP Department & Applied Scientist

Meteo-Algeria



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Research Interests

Numerical Weather Prediction systems development, atmospheric model physics and diagnostics, visibility and fog forecasting, microphysics parameterizations (LIMA scheme), software architecture for atmospheric sciences, operational meteorology automation, flood forecasting using coupled hydrometeorological models, scientific software engineering and opensource tool development.

Professional Experience

2020 à ce jour

Head of Numerical Weather Prediction Department

Meteo-Algeria (Feb 2024 - Present)

Leadership & Research Management

- **Strategic & Operational Leadership:** Directing the NWP department with a focus on team restructuring and organizational growth. Scaled the department to a multidisciplinary team of 13+ experts (Meteorologists, IT Specialists, and Research Engineers).
- **Vision & Strategy:** Architected and implemented a longterm NWP roadmap aligned with World Meteorological Organization (WMO) standards and national operational priorities.
- **Operational Excellence:** Coordinating the full lifecycle of forecasting suites, ensuring the seamless transition of high performance numerical models from R&D to 24/7 operational production.
- **International Scientific Engagement:**
 - **ACCORD Consortium:** Active in scientific activities, contributing to the design of advanced physical parameterizations and new numerical tools.
 - **ICAO Coordination:** Active member of the Data Management Group (DMG).
- **Research Supervision:** Spearheading applied research projects in highimpact domains.

Applied Scientist

Meteo-Algeria (Nov 2020 - Fév 2024)

- **NWP models Developer:** Actively engaged in the ACCORD Consortium, focusing on the continuous evolution of Numerical Weather Prediction models and advanced technical skills.
- **Physics & Microphysics Research:** Specialized in fog forecasting through indepth sensitivity studies of various microphysical schemes (including LIMA and ICE3).
- **System & Code Architecture:** Significant contributions to improving NWP code structure, with a strong focus on system portability, CMake build systems, and libraries externalizations
- **Operational Excellence:** Lead responsible for of the operational NWP suite at MétéoAlgérie.

Education

- **Master of Science in Hydrology, Climatology & Environment** 2021 2022

Université des Sciences et de la Technologie Houari Boumediene (USTHB), Algeria

Thesis: "Contribution of Dynamic Modeling to Flood Forecasting"

 - Implemented HYPE (Hydrological Predictions for the Environment) model for Algeria
 - Study area: Nothernn Algiers coastal basin (11 subbasins)
 - Created SoilLand Cover classes (SLC) from FAO, HWSD, GlobCover databases
 - Validation period: 19861995 observations from ANRH
 - Statistical metrics: NashSutcliffe Efficiency (NSE) scores

Supervisors: Prof. Amina SABRI (USTHB), Mr Abdenour AMBAR (ONM)
- **Engineering Degree in Meteorology** 2021 2022

WMO Regional Training Centre , IHFR , Algeria

Graduation Project: "Study of relationship between JenkinsonCollison weather types and aerosols"

 - Statistical analysis of atmospheric circulation patterns
 - ALADINClimate model Aerosols data processing and analysis
 - Lamb weather type classification application to North Algeria
 - Aerosolweather type correlation investigation

Supervisors: Mr Hakim ABANE , Mr Abdenour AMBAR (ONM)
- **Scientific Baccalaureate** 2021 2022

Lycée ElKhensa , Algeria

Research Stays & Collaborations

- **ACCORD Scientific Stay, CNRM/GMAP, MeteoFrance, Toulouse, France** 2024

Project: "Towards a more portable EPyGrAM (FA, ecTRANS, PyPI)"

 - Restructured EPyGrAM python package architecture using srclayout organization
 - Implemented setuptools configuration for PyPI distribution
 - Created falfilfa4py package: Python interface for FA/LFI/LFA formats from ACCORD.
 - Docker usage manylinux2014 container for crossplatform wheel generation
 - Published to PyPI: versions 1.5.0 (stable), 1.5.1 (development)

Supervisor: Alexandre Mary (CNRM/MeteoFrance)
- **ACCORD Scientific Stay, CNRM/GMAP, MeteoFrance, Toulouse, France** 2023

Project: "Externalization of FALFILFA formats from IFsARPEGELAM (IAL) Code"

 - Analyzed IAL code architecture and identified FA/LFI/LFA dependencies
 - Created LFIFA library with CMake/ecbuild build system
 - Compilation testing: Intel 2018.5.274, GNU 9.4.0, GNU 11.3.0
 - Platform validation: Ubuntu 20.04, 22.04, Belenos (MF HPC)
 - Integration with EPyGrAM: replaced libs4py.so dependency
 - GMKPACK integration: pack configuration
 - Modified fiat package: added ENABLE ECCODES option
 - Prepared for CY50T1 integration

Supervisors: Alexandre Mary, Ryad El Khatib (CNRM/MeteoFrance)

Professional Trainings

- **Executive Development Program, Higher Institute of Management and Planning (ISGP)** 2025
- Certificat of Professional Development for Corporate Executives
- **WISER MENA SeaSCAPE Seasonal Prediction Training, WMO RTC Egypte & UK Met Office** 2025
- Certificate in Seasonal forecasting fundamentals , production and verification
- **Numerical Weather Prediction Phase I, WMO RTC , Cairo, Egypt** 2021
- Certificate in operational NWP systems, data assimilation fundamentals, model configuration
- **Management Skills, Institut Selmane, Oran, Algeria** 2019
- Certificate in team management and project coordination
- **Human Resources Management Skills, Institut Selmane, Oran, Algeria** 2018
- Certificate in HR fundamentals and team development

Languages

- **Tamazight** : Native
- **Arabic**: Full professional proficiency
- **French** : Full professional proficiency
- **English** : Professional Working Proficiency

IT & Technical Skills

- **Operating Systems:** Expert Linux (Scientific environments); Windows (Professional proficiency).
- **HPC & Systems:** Familiar with High-Performance Computing (Belenos, Fennec).
- **Programming:** Python (NumPy, Xarray, Cartopy...), Fortran 90/95 (Model code), Shell (Bash), R, and CMake.
- **Softwares:** GIS (QGIS, ArcGIS), NWP Tools (EPyGrAM, MetView, GL), LaTeX, and Microsoft Office.
- **Other Technical Skills:**
 - Data Formats: Expert in FA, LFI, GRIB 1/2, NetCDF, HRIT , TIF and Zarr.
 - DevOps: Git/GitHub, Python packaging, and modular software architecture.
 - Modeling: Advanced configuration of AROME, ALADIN, and HYPE models.

Research/Software Projects

Meteo-Algeria

Operational Models Verification Framework :

- Automated computation of verification metrics using HARP package
- Realtime data processing and storage
- Handles 10GB+ daily data volume from multiple model sources
- Interactive web interface for realtime score visualization
- Includes TAF (Terminal Aerodrome Forecast) visualisation module from an adapted version of HONGKONG TAF Control Application

Meteorological Fire Weather Risk System :

- Implemented HDW & Revised IFM (Developped by another colleague) fire danger index.
- Developed Firedz Python package for index calculation and visualization
- Automated daily map generation for operational distribution.
- Supervised the developpement & operational deployment of an interactive visualisation + bulletin share WebApp for fire risk management.

Satellite Products:

- Modelisation of Convective Rainfall Rate (CRR) estimator from satellite imagery (MSG/SEVIRI).
- Operational integration for nowcasting applications as an interactive WebApp.

Operational Systems Engineering:

- Developement of 02 Python Packages (Production Ready):
 - **PyTAPS** (Python Toolkit for Algerian Production Suite): Functions for Meteo-Algeria production automation.
 - **Firedz**: Fire weather indexes package traitement & visualisation (IFM & HDW)
- Both designed for pip installation and modular use

Data Infrastructure :

- **Implementation of Zarr format as a based database architecture for model outputs**
 - Designed for machine learning applications
 - Efficient storage and access of multidimensional atmospheric data
 - Ready to visualise with xarray and cartopy.

Model Configuration & HPC:

- AROME configurations: multiple resolutions and vertical levels
 - Running on Algeria HPC and MeteoFrance R&D HPC
 - ARPEGE(Global) & ALADIN(LAM) models coupling.
 - Cycle compilation: CY43, CY46, CY48, CY50 of IAL Code.

Flood Forecasting System:

- Implementation of HYPE hydrological model for research purposes :
 - Configuration for Algiers coastal basin (11 subbasins, 4 provinces)
 - Coupled with ERA5 For a Warm Start Then AROME model for hydrological forecasting.
 - NashSutcliffe efficiency: 0.58 (Mazafran), 0.50 (ElHarrach)
 - Validated against 11 year observation period (1986-1995) and flood event (March 1985)